

Trios Cyber Internship — Day 27 Report

Mini Recon Workflow

Date: 27 September 2026
Target: 127.0.0.1
Environment: Kali Linux local lab
Status: Completed

1. Objective

The objective of Day 27 was to perform a basic reconnaissance workflow against one authorized lab target. The workflow included connectivity verification, Nmap service enumeration, web technology and HTTP header review, and directory discovery using FFUF. All reconnaissance activity was performed against the locally hosted target 127.0.0.1.

2. Lab Environment

Item	Details
Operating System	Kali Linux
Target	127.0.0.1
Web Server	Apache HTTP Server
Web Server Version	2.4.68
Platform	Debian Linux
Tools	Ping, Curl, Nmap, WhatWeb, FFUF
Authorization	Local controlled lab environment

3. Connectivity Check

The target was first tested using ICMP:
““text ping -c 4 127.0.0.1 cat > Day-27-Mini-Recon/DAY-27-REPORT.md «‘EOF’ # Trios Cyber Internship — Day 27 Report

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Authorization	Local controlled lab environment

3. Connectivity Check

The target was tested using ICMP with `ping -c 4 127.0.0.1`.

The test produced 4 packets transmitted, 4 packets received, and 0% packet loss. The average round-trip time was 0.085 ms. The target was therefore reachable and suitable for further reconnaissance.

4. Nmap Service Enumeration

Nmap service-version detection was performed using `nmap -sV 127.0.0.1`.

The final scan identified the following service:

Port	State	Service	Version
80/tcp	Open	HTTP	Apache httpd 2.4.68 (Debian)

Nmap reported the host as up and identified HTTP as the only open TCP service among the default 1,000 scanned ports. The remaining 999 scanned ports were closed.

During the assessment, an initial Nmap scan was performed while Apache was stopped and consequently showed no open ports. Apache was then started and verified as active before the final Nmap scan was repeated. This demonstrated the importance of validating service availability before interpreting enumeration results.

5. WhatWeb Technology Review

WhatWeb was used to fingerprint the HTTP service at `http://127.0.0.1/`.

The results identified an HTTP 200 OK response, Apache version 2.4.68, Debian Linux, the HTTP server identification `Apache/2.4.68 (Debian)`, the IP address 127.0.0.1, and the page title `Apache2 Debian Default Page: It works`.

The root URL served the default Apache Debian page rather than an application-specific page.

6. HTTP Header Review

HTTP response headers were reviewed using `curl -I http://127.0.0.1/`.

The following headers were observed:

Header	Value
Status	200 OK
Server	Apache/2.4.68 (Debian)
Last-Modified	Mon, 06 Jul 2026 18:11:35 GMT
ETag	"29cf-655f5346cada2"
Accept-Ranges	bytes
Content-Length	10703
Vary	Accept-Encoding
Content-Type	text/html

The response confirmed that the web server was accessible and disclosed its Apache version and Debian platform through the **Server** header.

7. Directory Discovery with FFUF

FFUF was used for content discovery against `http://127.0.0.1/FUZZ` with the standard Kali DIRB common wordlist containing 4,614 entries.

The scan identified the following paths:

Path	Status	Size	Observation
/	200	10,703 B	Apache default page
/index.html	200	10,703 B	Default index page
/javascript	301	351 B	Redirect response
/server-status	200	4,352 B	Apache status endpoint

The `/server-status` endpoint was identified during content discovery and documented as a reconnaissance finding only. No further interaction or exploitation was performed.

The scan processed all 4,614 entries and reported 0 errors.

8. Consolidated Reconnaissance Summary

Category	Finding
Target	127.0.0.1
Connectivity	Reachable, 0% packet loss
Open TCP port	80/tcp
Service	HTTP
Web server	Apache 2.4.68
Platform	Debian Linux
HTTP status	200 OK
Root page	Apache Debian Default Page
Discovered content	/index.html, /javascript, /server-status
FFUF entries tested	4,614
FFUF errors	0

The reconnaissance established that the local target exposes an Apache HTTP service on TCP port 80. Technology fingerprinting and HTTP header review confirmed the Apache/Debian environment. FFUF identified several accessible paths, including the Apache `server-status` endpoint.

9. Security Observations

The reconnaissance produced several useful observations. First, the HTTP service exposes the Apache version and operating-system platform through the `Server` header. Second, the default Apache page is accessible from the root path. Third, the `/server-status` endpoint is accessible and was discovered through content enumeration. Fourth, service enumeration results depend on the current state of the target service, as the initial scan showed no open ports because Apache was stopped. Finally, directory discovery can reveal additional web resources that are not immediately visible from the main page.

These observations are reconnaissance findings and do not by themselves establish a vulnerability.

10. Tools Used

Nmap: Used for service and version enumeration.

WhatWeb: Used for web technology fingerprinting.

Curl: Used for connectivity verification and HTTP header review.

FFUF: Used for directory and content discovery.

11. Evidence Files

Raw command outputs are stored in the `logs/` directory:

- `ffuf-directory-discovery.txt`
 - `http-headers.txt`
 - `nmap-scan.txt`
 - `whatweb.txt`
-

12. Evidence Screenshots

Screenshot 1 — Connectivity Check

Screenshot 2 — Nmap Service Scan

Screenshot 3 — WhatWeb Technology Review

Screenshot 4 — HTTP Header Review

Screenshot 5 — FFUF Directory Discovery

13. Safety and Authorization

All reconnaissance activity was performed exclusively against `127.0.0.1`, a locally controlled lab environment. No external systems or unauthorized targets were scanned. The `/server-status` finding was documented as part of reconnaissance and was not further exploited.

```
Session Actions Edit View Help
(yuvaraj@kali) ~
$ ping -c 4 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data:
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.164 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.085 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.039 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.054 ms
--- 127.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3070ms
rtt min/avg/max/mdev = 0.039/0.085/0.164/0.048 ms
(yuvaraj@kali) ~
$
```

Figure 1: Connectivity check

```
Session Actions Edit View Help
(yuvaraj@kali) ~
Host is up (0.0000020s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.27 seconds

(yuvaraj@kali) ~
$ sudo systemctl status apache2 --no-pager
[sudo] password for yuvaraj:
o apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)
   Active: inactive (dead)
   Docs: https://httpd.apache.org/docs/2.4/

(yuvaraj@kali) ~
$ sudo systemctl start apache2

(yuvaraj@kali) ~
$ sudo systemctl is-active apache2
active

(yuvaraj@kali) ~
$ curl -I http://127.0.0.1/
HTTP/1.1 200 OK
Date: Sun, 27 Sep 2026 16:21:12 GMT
Server: Apache/2.4.68 (Debian)
Last-Modified: Mon, 06 Jul 2026 18:11:35 GMT
ETag: "29cf655f5346cadaz"
Accept-Ranges: bytes
Content-Length: 10703
Vary: Accept-Encoding
Content-Type: text/html

(yuvaraj@kali) ~
$ nmap -sV 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-27 21:56 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache httpd 2.4.68 ((Debian))

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.24 seconds

(yuvaraj@kali) ~
$
```

Figure 2: Nmap service scan

```
Session Actions Edit View Help
Vary: Accept-Encoding
Content-Type: text/html

(yuvaraj@kali) [-]
$ nmap -sV 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-27 21:56 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache httpd 2.4.68 ((Debian))

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.24 seconds

(yuvaraj@kali) [-]
$ mkdir -p ~/Trios-Cyber-Internship/Day-27-Mini-Recon/Logs ~/Trios-Cyber-Internship/Day-27-Mini-Recon/screenshots

(yuvaraj@kali) [-]
$ nmap -sV 127.0.0.1 | tee ~/Trios-Cyber-Internship/Day-27-Mini-Recon/logs/nmap-scan.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-27 21:56 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache httpd 2.4.68 ((Debian))

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.33 seconds

(yuvaraj@kali) [-]
$ nmap -sV 127.0.0.1 | tee ~/Trios-Cyber-Internship/Day-27-Mini-Recon/logs/nmap-scan.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-27 21:57 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache httpd 2.4.68 ((Debian))

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.36 seconds

(yuvaraj@kali) [-]
$ whatweb http://127.0.0.1/ | tee ~/Trios-Cyber-Internship/Day-27-Mini-Recon/logs/whatweb.txt
http://127.0.0.1/ [200 OK] Apache[2.4.68], Country[RESERVED][XX], HTTPServer[Debian Linux][Apache/2.4.68 (Debian)], IP[127.0.0.1], Title[Apache2 Debian Default Page: It works]

(yuvaraj@kali) [-]
$
```

Figure 3: WhatWeb technology review

```
Session Actions Edit View Help
(yuvaraj@kali) [-]
$ mkdir -p ~/Trios-Cyber-Internship/Day-27-Mini-Recon/Logs ~/Trios-Cyber-Internship/Day-27-Mini-Recon/screenshots

(yuvaraj@kali) [-]
$ nmap -sV 127.0.0.1 | tee ~/Trios-Cyber-Internship/Day-27-Mini-Recon/logs/nmap-scan.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-27 21:56 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache httpd 2.4.68 ((Debian))

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.33 seconds

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(yuvaraj@kali) [-]
$ whatweb http://127.0.0.1/ | tee ~/Trios-Cyber-Internship/Day-27-Mini-Recon/logs/whatweb.txt
http://127.0.0.1/ [200 OK] Apache[2.4.68], Country[RESERVED][XX], HTTPServer[Debian Linux][Apache/2.4.68 (Debian)], IP[127.0.0.1], Title[Apache2 Debian Default Page: It works]

(yuvaraj@kali) [-]
$ curl -I http://127.0.0.1/ | tee ~/Trios-Cyber-Internship/Day-27-Mini-Recon/logs/http-headers.txt
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload  Total   Spent    Left   Speed
0   10703    0     0    0     0      0      0  0  0  0  0
HTTP/1.1 200 OK
Date: Sun, 27 Sep 2026 16:30:20 GMT
Server: Apache/2.4.68 (Debian)
Last-Modified: Mon, 06 Jul 2026 18:11:35 GMT
ETag: "29cfc655f5346cad2"
Accept-Ranges: bytes
Content-Length: 10703
Vary: Accept-Encoding
Content-Type: text/html

(yuvaraj@kali) [-]
$
```

Figure 4: HTTP header review

```
Session Actions Edit View Help
Vary: Accept-Encoding
Content-Type: text/html

(yuvaraj@kali) ~
$ ls -lh /usr/share/wordlists/dirb/common.txt
-rw-r--r-- 1 root root 36K Nov 17 2014 /usr/share/wordlists/dirb/common.txt

(yuvaraj@kali) ~
$ ffuf -u http://127.0.0.1/FUZZ -w /usr/share/wordlists/dirb/common.txt -mc 200,204,301,302,307,308 -o ~/TrioS-Cyber-Internship/Day-27-Mini-Recon/logs/ffuf-directory-discovery.txt

v2.1.0-dev

:: Method      : GET
:: URL         : http://127.0.0.1/FUZZ
:: Wordlist    : FUZZ: /usr/share/wordlists/dirb/common.txt
:: Output file : /home/yuvaraj/TrioS-Cyber-Internship/Day-27-Mini-Recon/logs/ffuf-directory-discovery.txt
:: File format : json
:: Follow redirects : false
:: Calibration  : false
:: Timeout     : 10
:: Threads     : 40
:: Matcher     : Response status: 200,204,301,302,307,308

index.html      [Status: 200, Size: 10703, Words: 3427, Lines: 369, Duration: 167ms]
javascript     [Status: 200, Size: 10703, Words: 3427, Lines: 369, Duration: 2ms]
server-status  [Status: 301, Size: 351, Words: 21, Lines: 10, Duration: 0ms]
:: Progress: [4614/4614] :: Job [1/1] :: 0 req/sec :: Duration: [0:00:00] :: Errors: 0 ::
[INFO] ----- PAUSING -----

entering interactive mode
type "help" for a list of commands, or ENTER to resume.
>
[INFO] ----- RESUMING -----

:: Progress: [4614/4614] :: Job [1/1] :: 2 req/sec :: Duration: [0:01:25] :: Errors: 0 ::

(yuvaraj@kali) ~
$
```

Figure 5: FFUF directory discovery

14. Conclusion

The Day 27 Mini Recon Workflow was completed successfully. The authorized local target was verified as reachable, its HTTP service was enumerated with Nmap, web technologies and response headers were reviewed using WhatWeb and Curl, and accessible web paths were identified using FFUF. The results, raw command logs, screenshots, and consolidated analysis are included as part of the Day 27 evidence package.